TARGET HOSTING PLATFORM

We plan to deploy the system on **Amazon Web Services (AWS)** cloud infrastructure.

Why AWS?

- Scalable infrastructure
- Reliable email server connectivity
- Managed database services
- Security controls
- Industry standard cloud platform

PRESENTATION LAYER HOSTING

Deployment Option:

- Hosted on **Amazon EC2**
  OR
- Using **AWS Elastic Beanstalk** (for simplified deployment)

Configuration:

- Ubuntu 22.04 LTS
- Python (Flask / Streamlit)
- NGINX reverse proxy
- HTTPS enabled

Role:

- Hosts:
  - Admin Dashboard
  - User Login
  - Manual Override Interface
- Communicates with Business Layer via internal APIs

BUSINESS LOGIC LAYER HOSTING

Deployment:

- Separate EC2 instance for better scalability

Contains:

- Email Fetching Module
- Spam Detection Module
- Intent Classification Module
- Priority Classification Module
- Rule Engine
- Workflow Controller

Communication:

- Internal REST APIs
- Direct function calls

DATA LAYER HOSTING

Database:

- Hosted on Amazon RDS
- MySQL Engine

Stores:

- Emails
- User credentials
- Logs
- Templates
- Configuration rules

Security:

- RDS placed inside private subnet
- Only Business Layer can access DB
- No direct public access

DEPLOYMENT STRATEGY

1. Infrastructure Setup

- Launch EC2 instance
- Configure Security Groups:
  - Allow HTTP (80)
  - Allow HTTPS (443)
  - Allow SSH (22 – restricted)
- Create RDS MySQL instance

2. Server Configuration

- **Install:**
  - Python
  - Flask / Streamlit
  - Required ML libraries
  - MySQL connector
- Configure NGINX reverse proxy
- Enable **systemd** service for app auto-star

3. API Configuration

- Presentation Layer → Calls Business Layer APIs
- Business Layer → Connects to RDS using secure credentials
- Email Fetching Module → Connects to IMAP/SMTP (Gmail or enterprise server)

4. Database Initialization

- **Create tables:**
  - users
  - emails
  - logs
  - templates
- Configure indexing for performance

5. Continuous Deployment

- Code managed via GitHub
- CI/CD using GitHub Actions
- Auto-deploy on push

SECURITY MEASURES

Network Security

- AWS Security Groups (Firewall rules)
- RDS in private subnet
- Disable root login

Data Security

- HTTPS using SSL certificates
- Password hashing using bcrypt
- Encrypted DB connections

Access Control

- Role-Based Access Control (RBAC)
    - Admin
    - User
- JWT-based session authentication

Email Security

- Use App Password (not main email password)
- TLS encryption for SMTP/IMAP

SCALABILITY STRATEGY

Currently:

- Layered monolithic deployment

Future Upgrade:

- Separate EC2 for:
  - ML Processing
  - Email Fetching
- Transition to Microservices
- Use Auto Scaling Groups